S-BPM in Surgery Simulation Training

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ABSTRACT
Implementation of simulation technologies for surgery training is a challenging problem nowadays that involves modern information technologies to perform a breakthrough in medical higher education. Still using of various surgery simulators in practice requires much assistance from teachers that need to monitor the training process and manage the students. In order to reduce these efforts and improve the efficiency of simulation surgery training there is proposed a software solution based on subject-oriented representation of training process, which allows higher customization of education making it adaptive and individualized. This paper describes this solution based on the concept of management by conditions and implemented in training simulators for laparoscopy and endovascular surgery training delivered as a result of “Virtual Surgeon” and “Inbody Anatomy” projects. The proposed approach was implemented by a software development kit (SDK) that is destined to develop on its basis new training simulation solutions for surgery education. The results and their probation in practice are illustrated by the simulation centre established at Samara State Medical University.

CCS Concepts
• **Applied computing➝Education➝Interactive learning environments.**

• **Computing methodologies➝Modeling and simulation.**

Keywords
Individualized Simulation; Surgery Training; Management by Conditions; Virtual Surgery; SDK.

1. INTRODUCTION
Surgery education is a complex process that requires a lot of efforts of both students and teachers. During the studies at a medical university the students need to obtain deep theoretical knowledge and fail-safe practical skills. To meet this goal each student needs to start practical training at the earliest courses, which is a slightly limited opportunity due to high responsibility to treat real patients. This problem can be solved by implementing the most up-to-date information technologies, advanced visualization and artificial intelligence to simulate real patients and let students train surgery intervention with high variations and no risks. Rich capabilities of modern simulation technologies well known from game development industry allow highly realistic representation of human body parts with a possibility to interact with them in real time. At the same time the education of each surgeon should be highly individualized to personal features and different background. Therefore there should be provided a solution for personalized education of surgeons based on an extensive use of modern visualization and simulation technologies and business process management.

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Simulation should become attractive for students to encourage them to obtain new skills, at the same time the training suite should understand the gaps and weaknesses of the student individually and adapt the training procedure to provide stable and long term effect. In addition to it there should be designed an Internet platform that provides to its users an opportunity to share new simulation technologies and products in integrated information space and adapt them considering the personal features of each student.

To solve this problem there is proposed to implement a subject-oriented approach for business processes management (S-BPM), which conceives a process as a collaboration of multiple subjects organized via structured communication [1]. Students can be considered not only as the objects of educational process, but also as subjects that interact with the training suite getting individualized educational services and negotiate with each other to proactively adapt the training procedure to personal needs.

In this paper there is proposed a model for interaction of students, teachers, trainers and 3D models and scenes developers in integrated information space. As a solution there has been developed a specialized software development kit (SDK), which contains a number of components that can be used to implement a large variety of simulation solutions for surgery training. The proposed approach allows the developers of new training suites to better specify the requirements, concretize the scope, prepare effective tests and improve training efficiency. The basic challenge of this research is to identify an interactive adaptive mechanism using 3D modelling and simulation for subject-oriented computer-aided education.

2. EXISTING SOLUTIONS OVERVIEW
The problem of simulation-based surgery education efficiency is actively discussed in scientific community. Intensive application of simulation tools for surgery training at medical universities and specialized training centers requires implementation of the most up-to-date technologies in robotics, 3D modeling, electronics and software engineering. Still the combination of the most realistic visualization of surgery field and adequate haptical feedback of manipulators that simulate surgery instruments can appear to be not enough for effective educational process. At the same time some simple versions of training suites that are comparatively cheap and easy in use can give a significant improvement to a surgery training technique.

For example, in terms of educational outcome complex virtual simulators can appear to be less effective than simple models. The papers [2 – 4] target a problem of looking for a balance between the best simulation and effective utilization of available technologies in practice. The paper [5] summarizes recent developments, technologies and actual educational needs and postulates that the educational value of simulation and training technologies changes as a surgeon progresses through his (or her) training, and introduces the concept of virtually augmented environment for different types of training through continual educational process. Thus training should be organized adaptively and individually through interactive tasks. Scientific progress requires difference of educational technologies [6]. Up-to-date surgical training should use effectively all of simulation capabilities, game technologies, haptics, and virtual environments. [7] highlights the main advantages of virtual technologies in surgery: evolved and realistic human anatomy with normal and pathological conditions and a structured learning environment with controlled levels of difficulty.

Technical aspect can be characterized by an extensive use of emerging and innovative technologies to provide high realism of visual scene and force feedback. Among the most widespread areas of simulation for surgery training there are laparoscopy, laparotomy and endovascular diagnostics and surgery [8]. The major solutions in this area are based on visualization of 3D scenes that represent surgery fields for different cases and simulation of surgery intervention by means of specifically designed manipulators. The concept is pretty close to gaming simulation: the student has a certain situation described by visual model with predefined features and can perform a number of actions getting the response that simulates the real human body behavior.

Our experience in this area includes the “Virtual Surgeon” and “Inbody Anatomy” projects that were carried out in 2012 – 2014 and deliver a number of simulation training suites for endovascular and laparoscopy surgery [9, 10]. These products were successfully probated in several medical universities. During the process of their deployment and technical support there was gathered a remarkable feedback from professors and IT departments. There was identified a strong request to separate the training suites into a number of autonomous components that can be used by universities themselves to build own solutions and adapt them for specific educational programs. Students, trainers and teachers, simulation scenarios and 3D scene developers should be able to cooperate in real time to provide an interactive process of surgery training making it adaptive and individualized.

So despite a variety of solutions presented at the market, e.g. for laparoscopy and endovascular surgery training, there is a number of challenges of their application in practice. Firstly, there is no universal procedure of their integration into educational process. Secondly, the methodology of surgery training differs from one university to another. Thirdly, most of the existing training suites require tutor’s assistance and cannot adapt to student’s personality and consider the human factor.

To solve these problems there was developed a specialized software development kit (SDK) intended to be used by software and hardware developers and medical educational community to provide new simulation solutions for medical universities. Surgery education can be considered as a specific business process aimed to give the students theoretical knowledge and practical skills in the most effective way. This process should be individualised and adaptive to different educational strategies; therefore the concept of SDK implementation should be based on perspective technologies of business processes management.

Comparing to standard notations of business processes formalization S-BPM [11 – 13] appears to be the most applicable in the given context. SDK-based application of S-BPM for software development can be illustrated by several successful projects, including healthcare and education. The modeling paradigm uses five symbols to model any process and allows direct transformation into executable form.

3. S-BPM MODEL FOR SURGERY TRAINING

Let us consider the process of surgery training from the subject-oriented perspective. There are three major types of **subjects** involved into the process:

- Student: active subject that performs surgery intervention actions using specifically designed manipulators;
- Training exercise: as opposed to traditionally implemented scenarios as passive objects, the exercise should be implemented as an autonomous subject that can receive the information about the surgery intervention events in the form of messages and coordinate its internal behaviour adjusting the scenario complexity to the current student’s level;
- Teacher: this subject receives messages that describe the current training progress and thus gets the possibility to monitor and remotely manage the training process.

The following objects are used for simulation:

- the models of human bodies or laparoscopy instruments.

To capture and evaluate the results of training process there is proposed to implement the following formal model.

Each surgery case can be described by a certain clinical picture and a disease pattern and has a corresponding 3D physical model simulating a human body for this case. Initially the student knows nothing about the current state except the clinical picture description and needs to perform a surgery intervention. Any mistake will cause damage and be visualized, so in case it can be mitigated the additional actions should be carried out.

For the considered case c_j, where $j = 1..N_c$ is the case number there can be defined the consecutive order of necessary and obligatory actions, represented by Boolean variables

$$e_{j,k}\left(c_j, v_{j,k}, t_{j,k}\right) \in \{0,1\}, \tag{1}$$

where the action $v_{j,k}$ should be ideally performed with average time interval $\Delta t_{j,k}$ after the previous action (or intervention start for the first one). Please note that the consequence $e_{j,k}$ does not specify the strict order of actions, but gives a certain number of milestones that are specific for the concerned case.

The factual surgery intervention training process can be formalized by the consecutive order of actions, represented by Boolean variables:

$$e'_{i,j,l}\left(u_i, c_j, v'_{i,j,l}, t'_{i,j,l}\right) \in \{0,1\}, \tag{2}$$

where u_i is the user that performs actions $v'_{i,j,l}$ at the corresponding moments of time $t_{i,j,l}$ from the intervention start.

$l = 1..N'_v$, and N'_v is the number of actions.

The student's mark given by a surgery simulator is usually a grade calculated on the basis of a number of key performance indicators (KPI) characterizing intrinsic factors like human body damage and blood loss and efficiency metric, which is evaluated on the basis of agility, length of the instruments' movement track and intervention duration. In order to introduce the features of game mechanics the overall grade can be calculated in the form of a score summarizing the evaluation of student's efforts and giving him an opportunity to improve his skills.

In addition to this we propose to specify one more grade that characterizes the variation of student's actions from the original intervention scenario:

$$\rho_{i,j} = \sum_{k=1}^{N_v} \Big(e_{j,k}\left(c_j, v_{j,k}, t_{j,k}\right) \cdot t_{j,k} - \\ - \sum_{l=1}^{N'_v} e'_{i,j,l}\left(u_i, c_j, v'_{i,j,l}, t'_{i,j,l}\right) \cdot t'_{i,j,l} \cdot \delta_{i,j,k,l} \Big), \tag{3}$$

where $\delta_{i,j,k,l} = \begin{cases} 1, v_{j,k} = v'_{i,j,l} \\ 0, v_{j,k} \neq v'_{i,j,l} \end{cases}.$

The statement (3) allows calculating the total deviation in time between the moments of expected actions and the moments of factual actions executed by the student. This KPI can be utilized either as a component of the summarizing grade, or as a separate indicator used by the training system to adapt the complexity of the scenario to the current student. It generalizes the idea of game mechanics being applied to the process of training based on simulation. It is proposed to adapt the complexity of simulation to the student's intermediate success.

For example, in case the student is novice to work with manipulators and surgery scene he will spend much time on each action, even in case he is good in theory and knows the correct procedure. Using the statement (3) the system will identify it and simplify the case (by e.g. extending the accuracy of manipulators). In reverse for a student with enough experience of operating with manipulators the statement (3) will present negative values, and the system will also recognize the situation. In this case there can be introduced additional random events (like e.g. unexpected effusion of blood). The ways of simplification and types of unexpected events can be easily specified for certain surgical cases.

4. SOLUTION VISION

To implement the principles of subject oriented business process management of surgery simulation based education we propose the solution illustrated by Fig. 1. Surgery training suite software provides a set of training exercises on the basis of surgery case and scenarios description, 3D models, and estimation (evaluation) technique.

Users perform the exercises using the components of training suite software and hardware. In addition to this regular functionality we propose to introduce 2 additional modules for identification of students' expertise on the basis of their faults retrieved from the training log and adaptation of exercises in real time. Such an approach provides a subject-oriented procedure of interactive technologies application in simulation surgery training. Identification of students' expertise allows considering the human factor and adapting the complexity of scenarios for different skill level of students, different types of learning and cognitive abilities, speed of training etc. Therefore training suite should conform to the student's skills level. It could force him to raise his skills interactively.

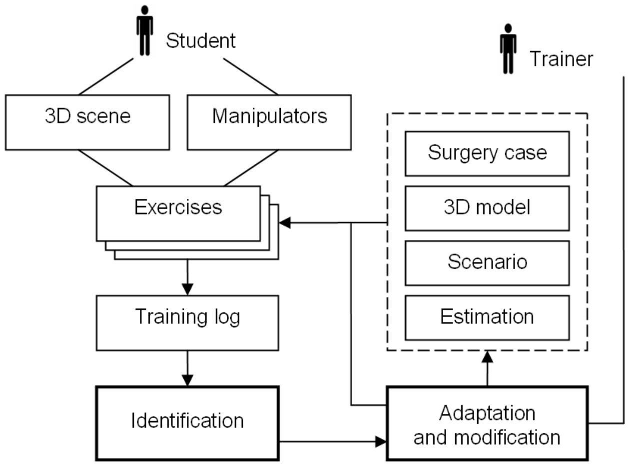

Figure 1. Personalized simulation concept for surgery training.

The main problem of simulation surgery training can be stated as the most effective utilization of interactive technologies to provide fast and stable learning. This can be formalized by two objectives: minimum average learning time with limited deviation and good average evaluation results (high grades estimation) just after the simulation and after certain check time interval.

It should be noted that simulation technologies can be used at different stages of surgery education: for students' introduction and basic skills training, and for advanced training and skills improvement of experienced doctors. Besides, the simulation training suites can be used together with new surgery equipment for familiarization. Due to the difference in use cases and target user groups as well as the distinction in professional skills and theoretical background of students the training scenarios and estimation procedures should be individualized. As soon as it is impossible to determine the students' level before the training the simulation scenarios should be adapted in real time.

According to this approach the consequence of student's actions in two different episodes of a certain training case can be significantly different. As a result the studying cases become not predetermined: the student knows nothing about the current state except the clinical picture description and needs to perform a surgery intervention.

To represent and describe the surgical cases in laparoscopy and endovascular simulators there was proposed to use LUA programming language. This has made it possible to separate code from the knowledge about the user's behavior, specify possible deviations from the standard surgery procedure and give the user an opportunity to fix the mistakes.

For example, in case the student is novice to work with manipulators and surgery scene he will spend much time on each action, even in case he is good in theory and knows the correct procedure. The system will identify it and simplify the case (by e.g. extending the accuracy of manipulators). In reverse for a student with enough experience of operating with manipulators there can be introduced additional random events (like e.g. unexpected effusion of blood). The ways of simplification and types of unexpected events can be easily specified for certain surgical cases.

Such an approach allows obtaining the following benefits:

- the training suite becomes user friendly and adapts to different students with various experience and entry level;

– the simulated cases attract attention and interest of students in real time;
– the complexity of cases increases accordingly the student's progress in training;
– the monotonicity of training decreases by means of incorporation of random component to the case script;
– during the process of simulation the fragments of scenarios that are familiar and therefore boring are reduced;
– the system combines the periods of getting pleasure from study and satisfaction from the achievements, which is critical for a good training suite.

5. IMPLEMENTATION FEATURES

At programming level the proposed implementation features can be illustrated by the following example of human body parts behaviour for laparoscopic cholecystectomy. Let us consider a gall bladder with a possibility of bleeding simulation. The links between the code samples are outlined with bold.

1. Events that are sent for processing to different scripts can be introduced for different groups of objects or a scenario. Soft bodies events are specified as the following:

```
a) bool SoftBody::startBleedingAtVertex(size_t
index){
size_t bleeding_id;
bool res = this->mesh_renderer-
>createWaterMapSourceAtVertex (index, 5, 5, 255,
&bleeding_id);
if(res==true){
Message m;
m.str_type = "ORGAN_BLEEDING_STARTED";
m.params["bleeding_id"] =
static_cast<float>(bleeding_id);
m.params["bleeding_rate"] = 255.0f;
m.id = this->getLastMessageId();
this->dispatchMessage(m);
}
return res;
}
b) void SoftBody::stopBleedingAtTextureCoordinate(
plTexCoord tex_coord, float uv_space_spot_size) {
std::unordered_set<size_t> stopped_bleedings;
this->mesh_renderer-
>removeWaterMapSourcesWithinSpot(tex_coord,
uv_space_spot_size, &stopped_bleedings);
for(size_t bleeding_id : stopped_bleedings) {
Message m;
m.str_type="ORGAN_BLEEDING_STOPPED";
m.params["bleeding_id"] =
static_cast<float>(bleeding_id);
m.id = this->getLastMessageId();
this->dispatchMessage(m);
}
}
```

2. The objects in scene are specified as the following. Different parameters can be specified like allocation, physical constants and options, references to other files, including the object processing script:

```
<softbody name="Gallblader">
<position x="0" y="0" z="0"/>
<script
filename="../resources/media/…/Gallbladder.lua"/>
<option type="vector3"
name="external_acceleration" value="0 0 -150"/>
<option type="bool" name="needs_surface_collide"
value="true"/>
<option type="string" name="asset"
value="LHE_NmlGlbd_Base_Gallbladder"/>
<option type="number" name="water_map_resolution"
value="512"/>
<option type="number" name="collision_group"
value="4"/>
<option type="float" name="tension_drop_limit"
value="3"/>
<option type="float" name="no_damage_penetration"
value="15"/>
<option type="float" name="max_damage_penetration"
value="20"/>
</softbody>
```

3. Events processing in LUA scripts are specified with links to physical entities that refer to a single object:

```
Gallbladder.lua
preprocessing = {
["ORGAN_BLEEDING_STARTED"] = function(msg)
SurfaceBleeding.addBloodSource(msg:GetParam("bleed
ing_id"),    msg:GetParam("bleeding_rate")/255    *
max_bleeding_amount)
end,
["ORGAN_BLEEDING_STOPPED"] = function(msg)
SurfaceBleeding.removeBloodSource(msg:GetParam("bl
eeding_id"))
end,
["COAGULATOR_BURNED"] = function(msg)
Util.displayMessage("Coagulator
burned"..msg:GetParam("body_injury"))
m = Message();
m.sender = this:GetName();
m.receiver = "EngineCore";
m.type = "INSTRUMENT_INJURY";
m:SetParam("amount", msg:GetParam("body_injury"));
this:SendMessage(m);
end
}
```

6. SURGERY TRAINING SUITE AND SDK

The described above principles were implemented in laparoscopy training suite described below.

3D models of human body parts set up the input for visualization core components that can generate laparoscopic or endovascular scenes close to real images processed in the process of surgery intervention. There were designed, developed and conjoined up to 3000 models of human body parts (see Fig. 2) combined to 12 layers of human body, including the ligaments, blood vascular system, innervations system, outflow tracts, lobar and segment structures of internals.

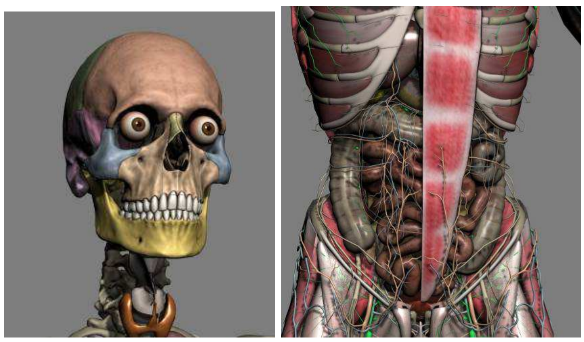

Figure 2. Anatomy 3D human body models.

To develop the models there were used real digital volume computer tomography and magnetic resonance tomography images. There was also delivered a shell viewer and a database of 3D models of human body (see Fig. 3) and is currently available in the market under the product trademark "Inbody Anatomy".

Figure 3. "Inbody Anatomy" product in use.

The described above 3D models of human body parts were used to develop the scenes for laparoscopy training (see Fig. 4). To provide realistic picture these models were fashioned with the help of specifically designed shaders and some fragments like jars, liquids and blood were implemented in software. These shaders and algorithms were implemented apart from the models; and any new models including those that are implemented by third party developers can be used to generate surgery scenes as well.

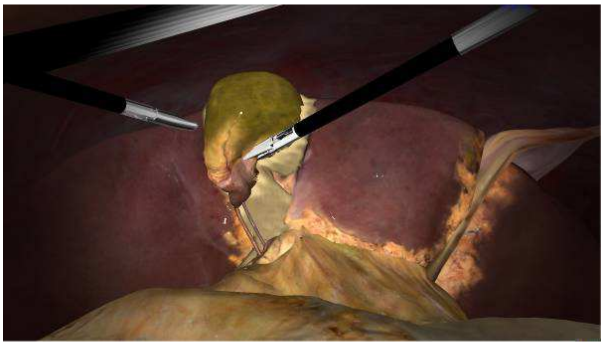

Figure 4. 3D Scene for laparoscopic cholecystectomy.

To provide highly realistic visual and physical models there were used Nvidia PhysX and Bullet physics library middleware. The inner parts of a human body are simulated in the scene by soft body models and surgery instruments are simulated by rigid bodies. The value of the feedback force is calculated on the basis of current geometrical position of soft and rigid bodies in the scene considering their deformation and/or topology distortion. The software was developed in C++ using .NET C# as a platform for user interface and infrastructure support.

The laparoscopic manipulator was built using 4 Dynamixel actuators that provide four degree-of-freedom feedback and movement. The main feature of the construction is that the entire outfit is integrated into the manipulator. This helps to reduce any transmission and as a result increases robustness and feedback sensitivity as compared with widely spread analogs. Such a solution entailed a necessity to develop specific algorithm of feedback force calculation based on transition of the physical model to forces and positions simulated by manipulators. The developed laparoscopic training suite use is presented at Fig. 5.

To provide high flexibility of the developed technologies and enforce its practical use there was developed SDK architecture, which contains a number of components that can be used to implement a large variety of simulation solutions for surgery training. SDK contains the components to adjust the educational scenarios, a universal module platform, and cloud services for technical support and professional communications.

Figure 5. Laparoscopy training suite in use.

This architecture forms the base of new simulation training suites. Most of the already released training suites and the ones that are planned to be delivered in the nearest future use the same approach of physical simulation, 3D visualization and force feedback manipulation. Therefore, these common modules should be separated and implemented with a high degree of universality of applications. As every up-to-date information system the training suites support distributed architecture with cloud services and local information environment build on top of data exchange system.

7. PERSPECTIVES OF S-BPM IN MEDICAL APPLICATIONS

Orientation to individualized and personalized educational technologies is a challenging area of possible application of subject-oriented business process management.

The proposed intelligent approach to capture the student's performance and adapt training scenarios according to his results allows developing a number of training suites capable of individual education. Other possible applications of subject oriented business processes management include various simulation software and hardware suites for realistic representation of 3D scenes and their interactive analysis.

Fig. 6 presents the extended version of 3D Atlas "Inbody Anatomy" that can be deployed on interactive board and used for extensive study of human body parts in a group under the experienced teacher's mentoring. Using the interactive technologies in addition to realistic simulation provides simple but effective educational facilities that can be used at different stages of anatomy and medical education: from higher schools to universities.

Fig. 7 presents another solution based on utilization of Virtual Reality (VR) headset using head mounted display for surgeon decision making support in the process of surgery intervention. The system called “Autoplan” is being developed for automated control and scheduling of surgery intervention. Intelligent scheduling is performed on the basis of personalized 3D models of human body derived from individual digital volume computer tomography and magnetic resonance tomography images.

Figure 6. Interactive board used for surgery education.

Figure 7. Augmented reality surgery training suite implemented using 3D surgery simulation SDK.

The system provides the unique functionality of decision making support that can be used for surgery simulation and control in real time that helps reducing the time of intervention and risks of postoperative complications.

8. CONCLUSION

Identification of users’ expertise and the corresponding adaptation of training scenarios make the simulated surgery intervention fascinating, i.e. attract the student’s interest in the process of acting. S-BPM concept is effective for formalization and modeling of this approach in practical applications. Surgery exercises can be performed without trainer’s assistance and the average learning time is minimized with the same learning effect.

9. ACKNOLEDGEMENT

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